6.10.2 Highway Lighting

The highway lighting provides visual conditions for safe, quick and comfortable movement of road users. Highway lighting are provided at intersections, level crossings, bridge sites and at places where there is restriction to traffic movements.

(i) Lighting Layouts

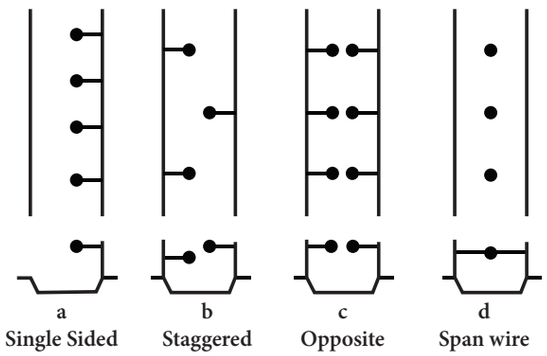
Depending on the road category there are various arrangements for two way traffic roads, namely

a. **Single sided lighting** – located on one side, if width of the road ≤ mounting height. Luminance at the opposite remote end is lower than under the lamp.

b. **Both side lighting - Staggered** – located on either side of the road in a staggered or zigzag fashion when width is 1 –1.5 times the mounting height. Here care is to be taken to avoid dark patches.

c. **Both side lighting - Opposite** – located opposite to one another. When width is greater than 1.5 times the mounting height.

d. **Span wire** – luminnaires suspended along the axis of the road for narrow roads only. Suspended from cables strung between buildings.



Lighting arrangement for 2 way Street

(ii) Design factors of Highway lighting

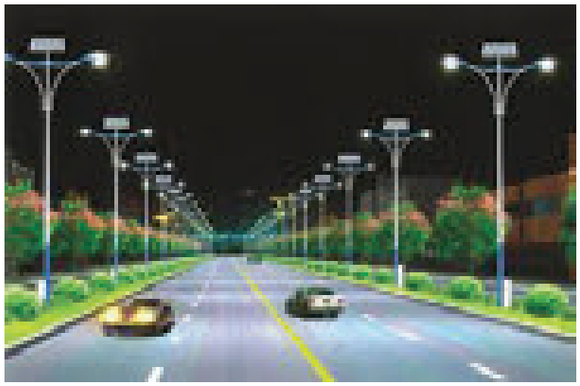
Following are the eight important factors to be considered in the design of highway lighting

1. Contrast
2. Glare
3. Lamps
4. Lateral placement of lighting poles
5. Lighting layouts
6. Luminaire distribution of light
7. Mounting height and overhang
8. Spacing of lighting units.

(iii) Benefits Of Highway Lighting

The various benefits of highway lighting are listed below

1. Reduces the risk of night time accidents.
2. Assist in the protection of property.
3. Making residents feel secure.
4. Illegal and anti-social activities on roads can be avoided.
5. Enhance the appearance of the area after dark.



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